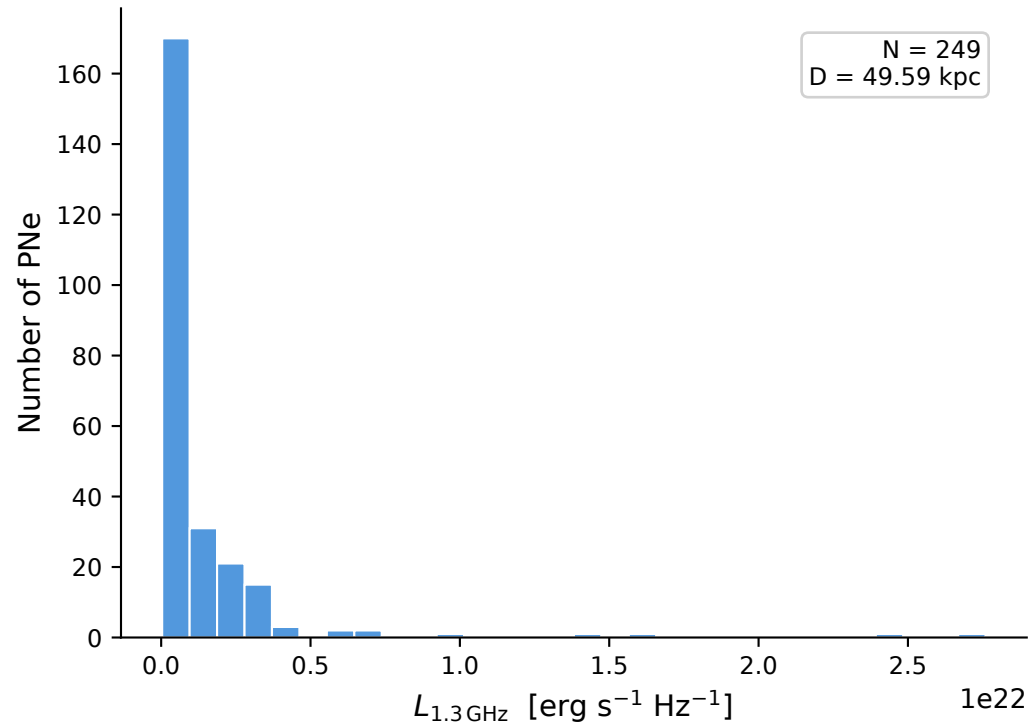
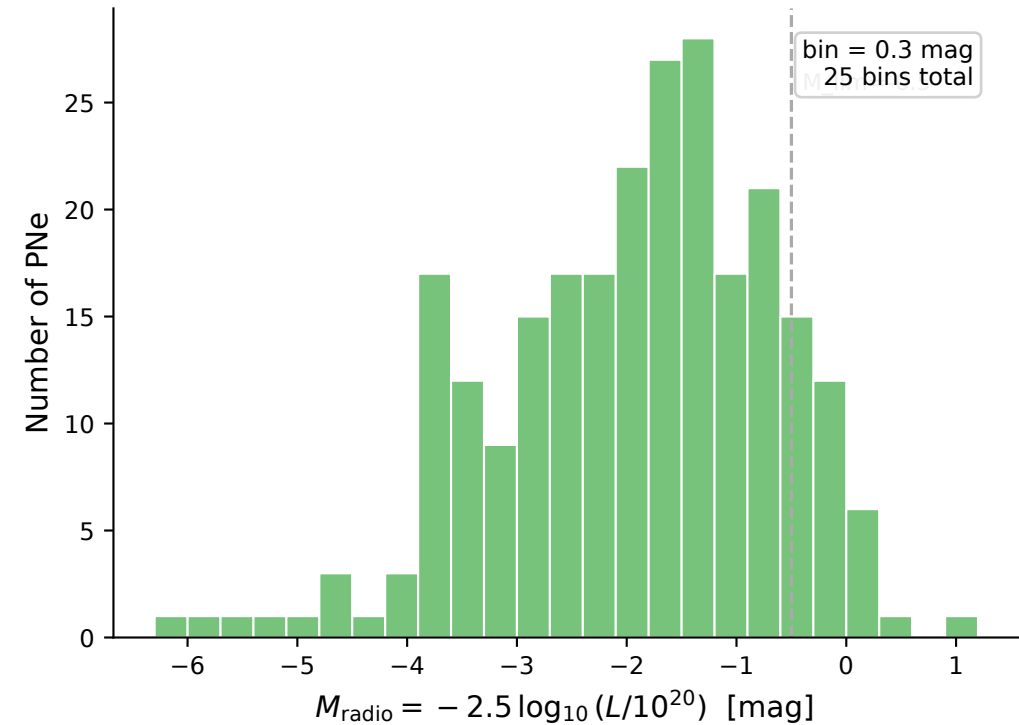


Data Preparation: From Luminosity to Fit-Ready PNLF

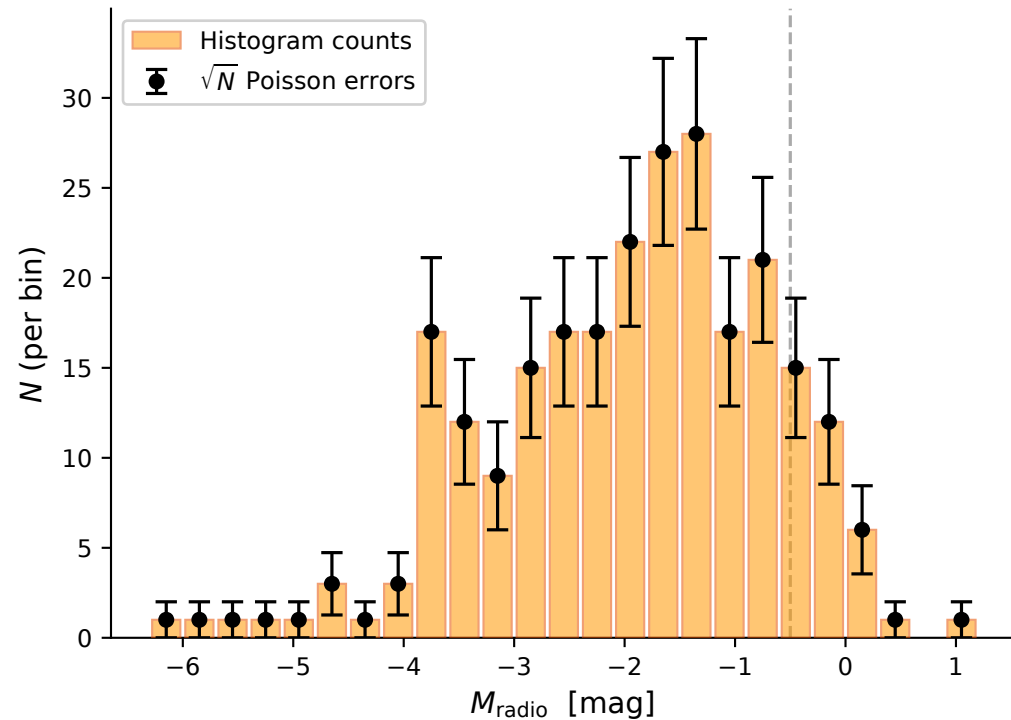
(a) Luminosity distribution



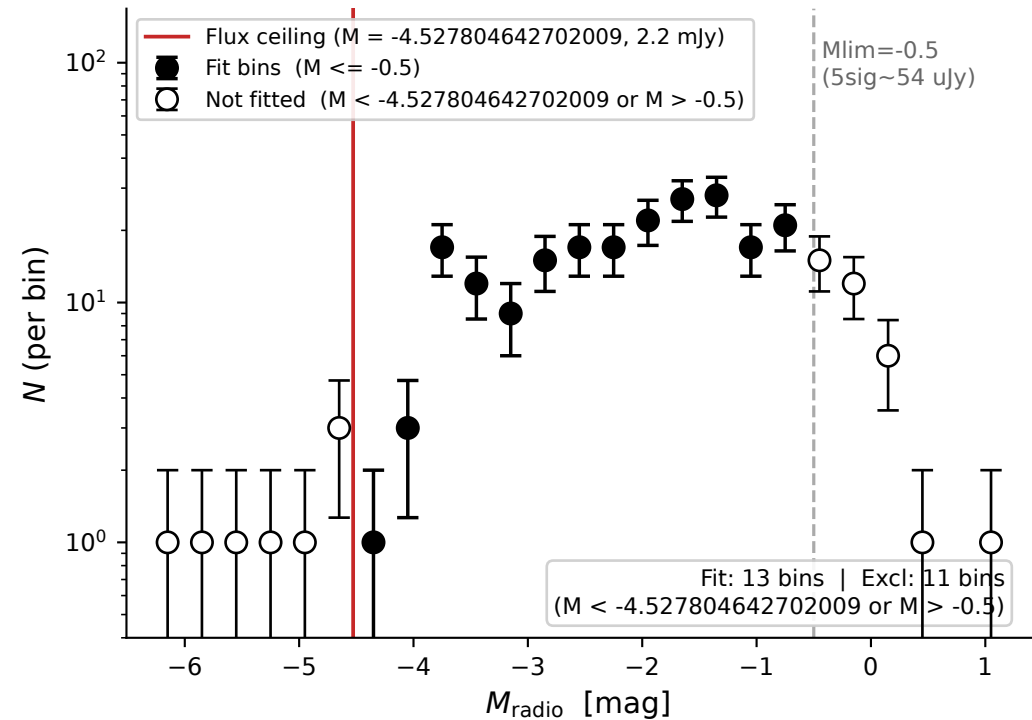
(b) Magnitude distribution

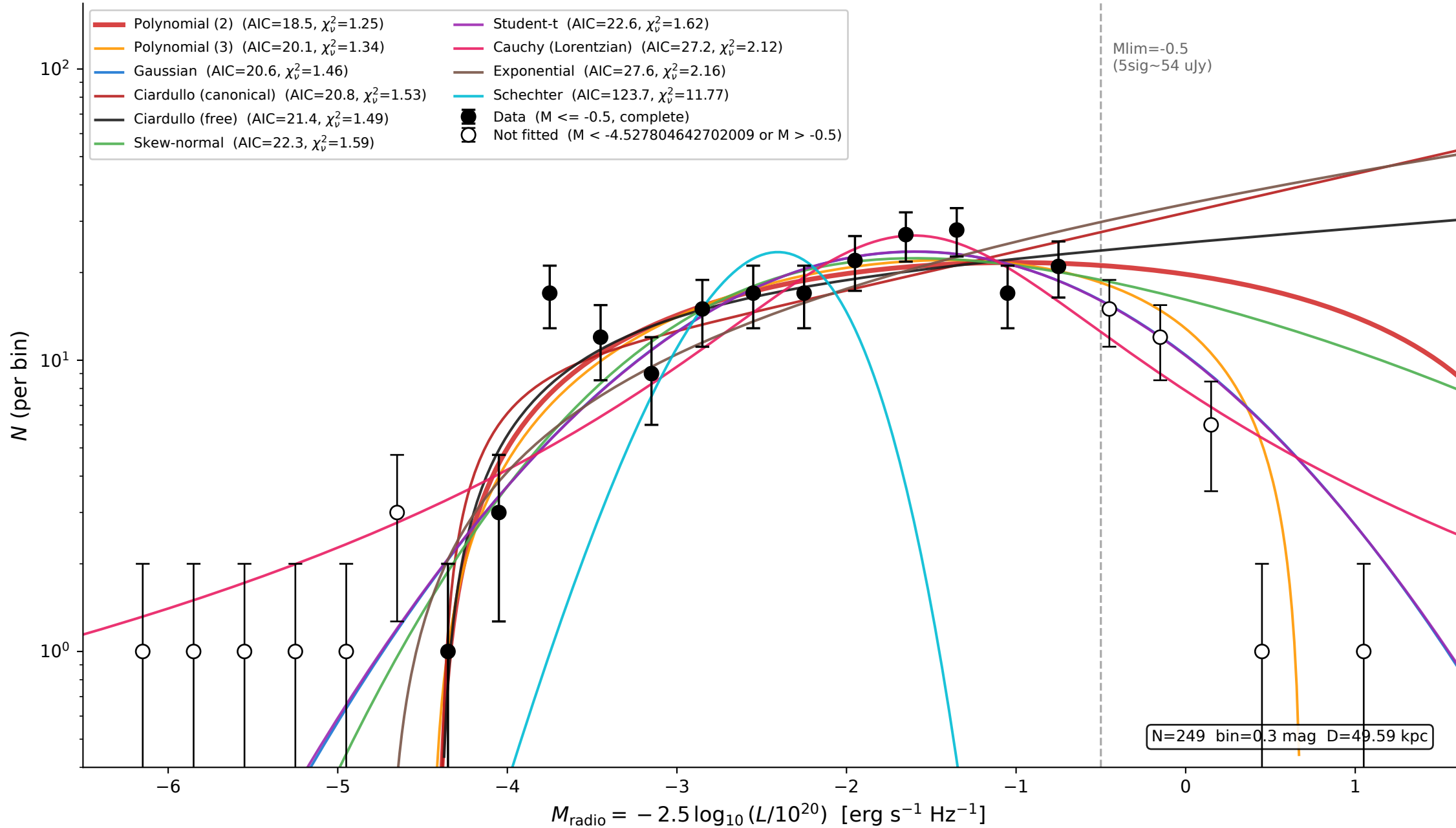


(c) Binned counts with Poisson uncertainties

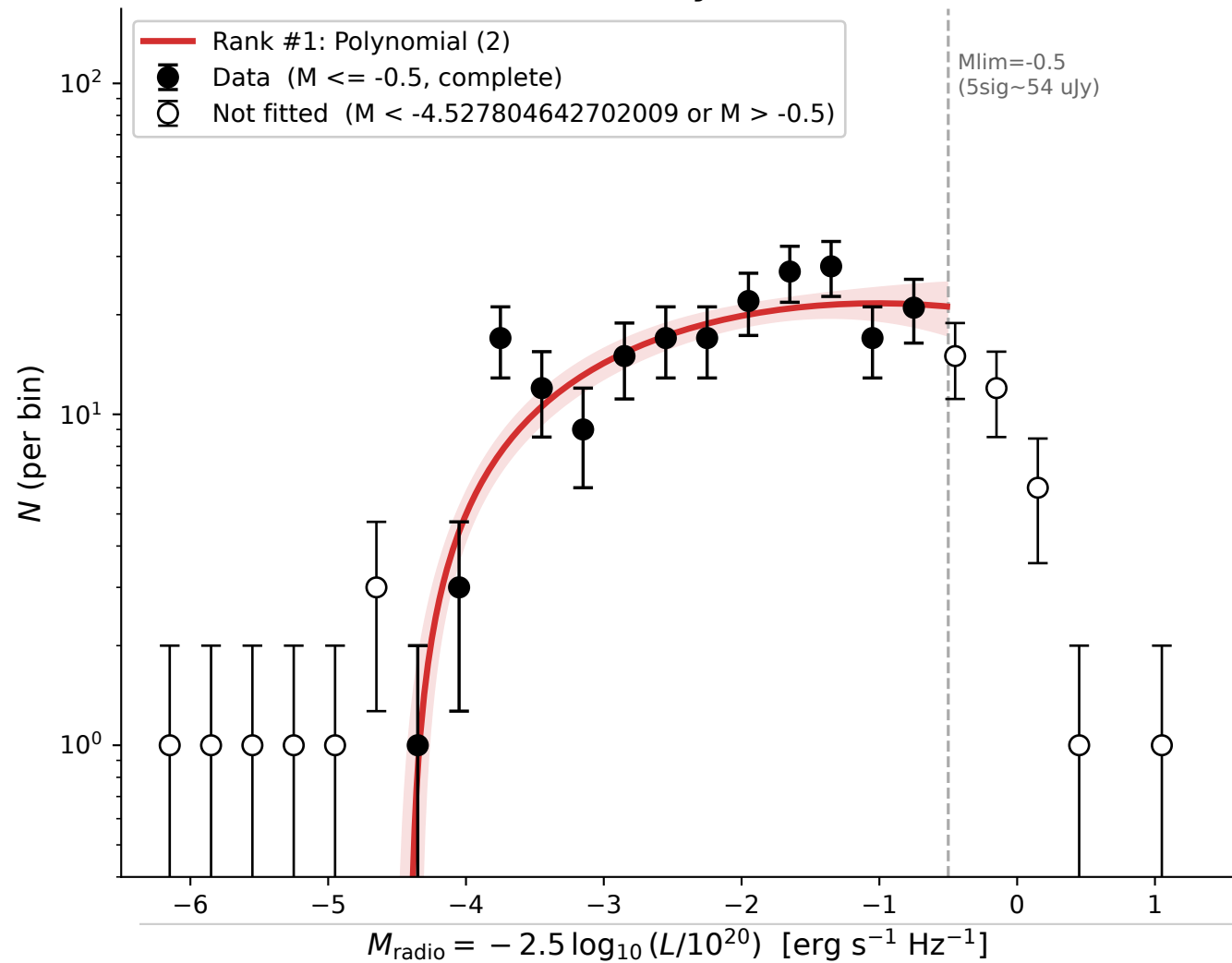


(d) Fit-ready data: complete vs excluded





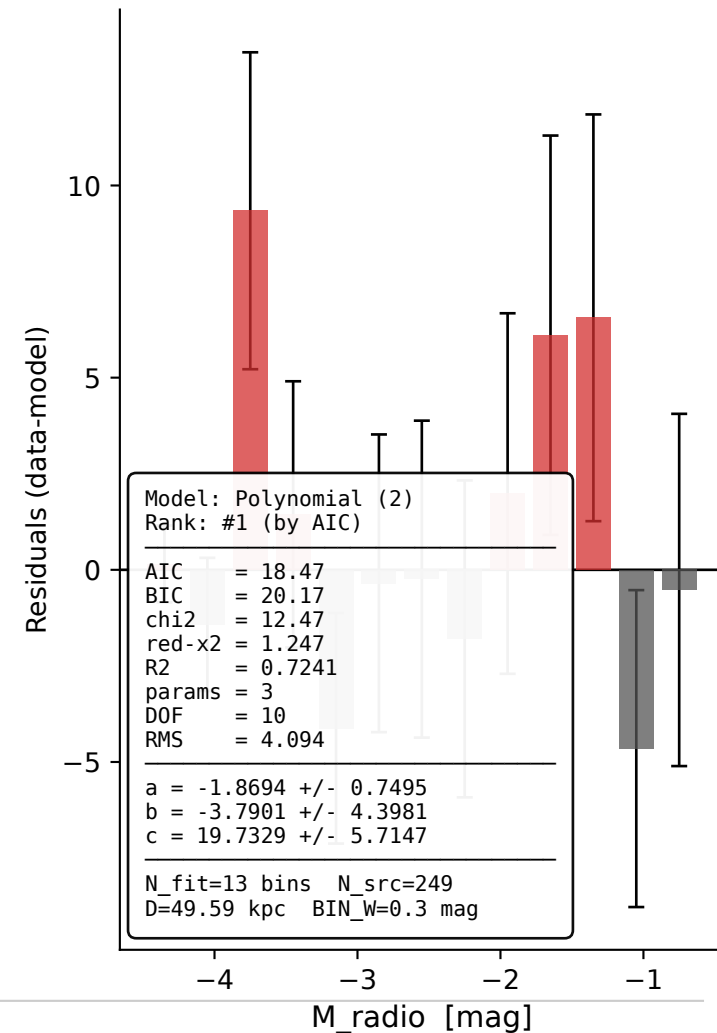
Rank #1: Polynomial (2)



General model equation:

$$N(M) = a \cdot M^2 + b \cdot M + c$$

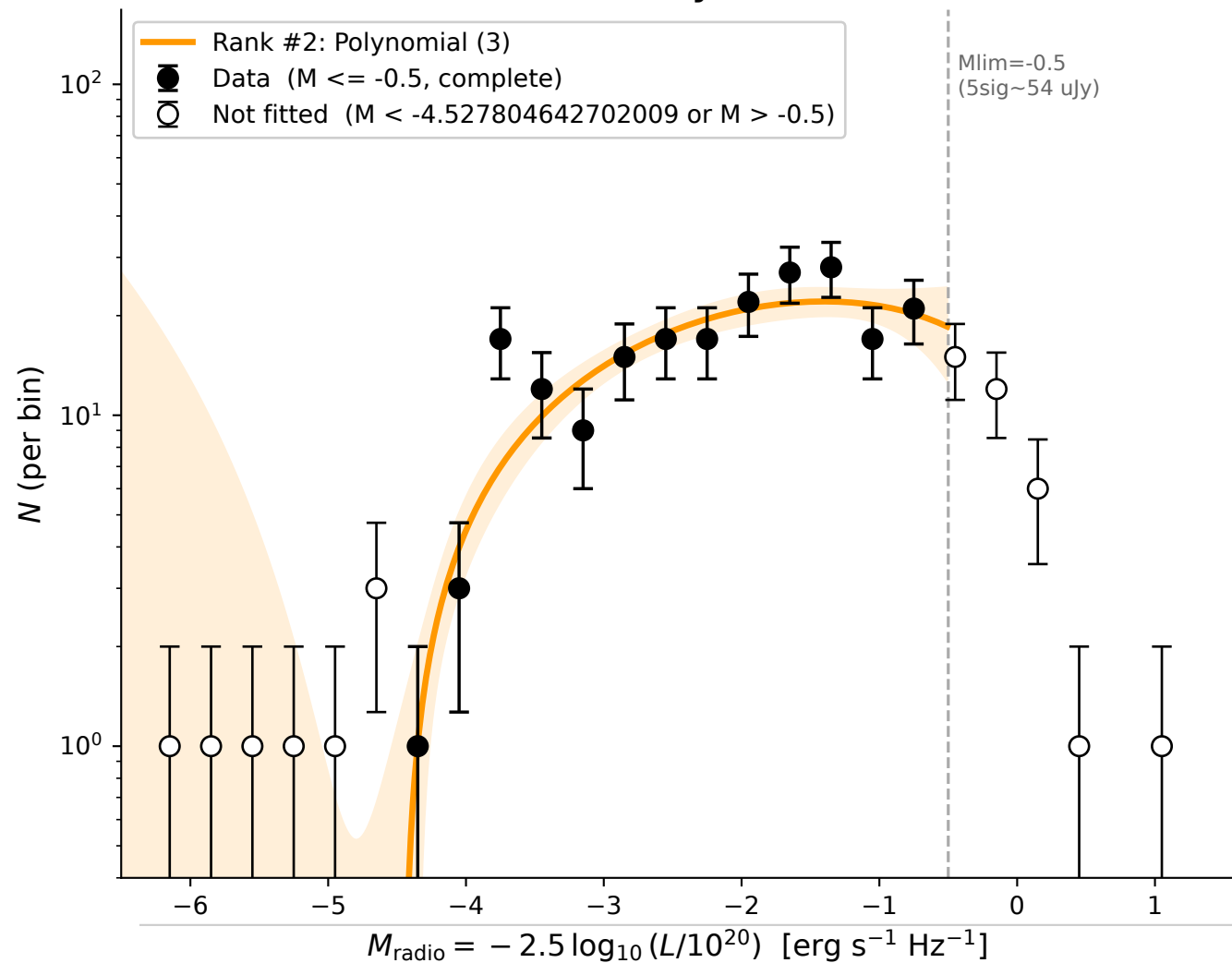
Residuals



Fitted equation (parameter values substituted):

$$N(M) = -1.8694 \cdot M^2 - 3.7901 \cdot M + 19.7329$$

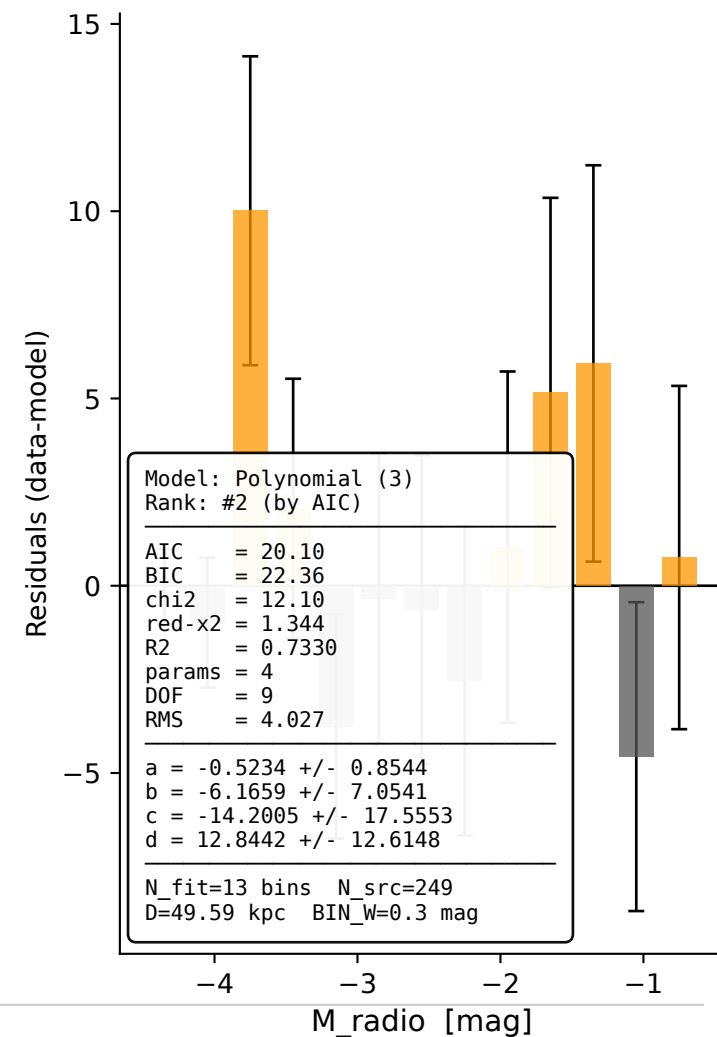
Rank #2: Polynomial (3)



General model equation:

$$N(M) = a \cdot M^3 + b \cdot M^2 + c \cdot M + d$$

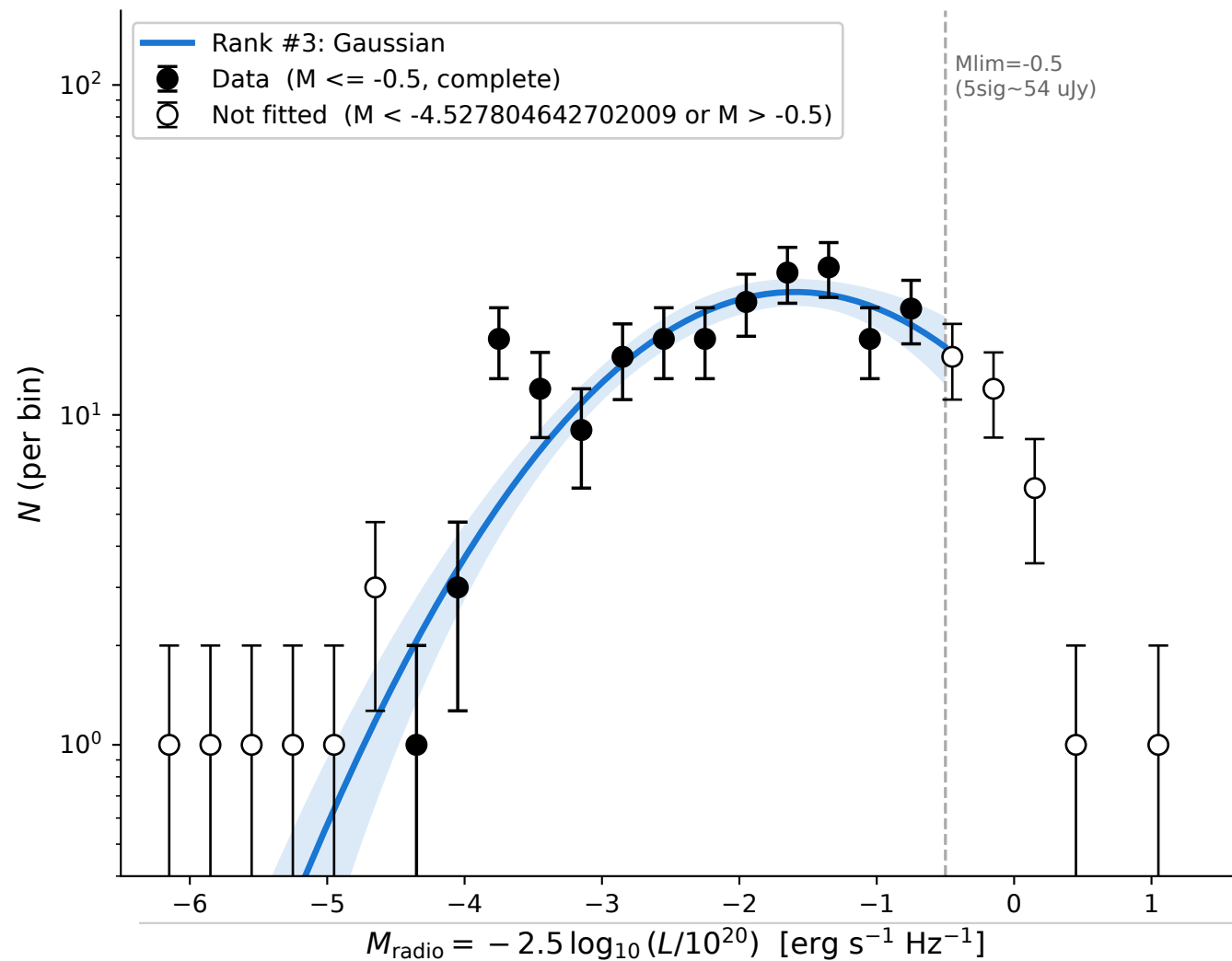
Residuals



Fitted equation (parameter values substituted):

$$N(M) = -0.5234 \cdot M^3 - 6.1659 \cdot M^2 - 14.2005 \cdot M + 12.8442$$

Rank #3: Gaussian



General model equation:

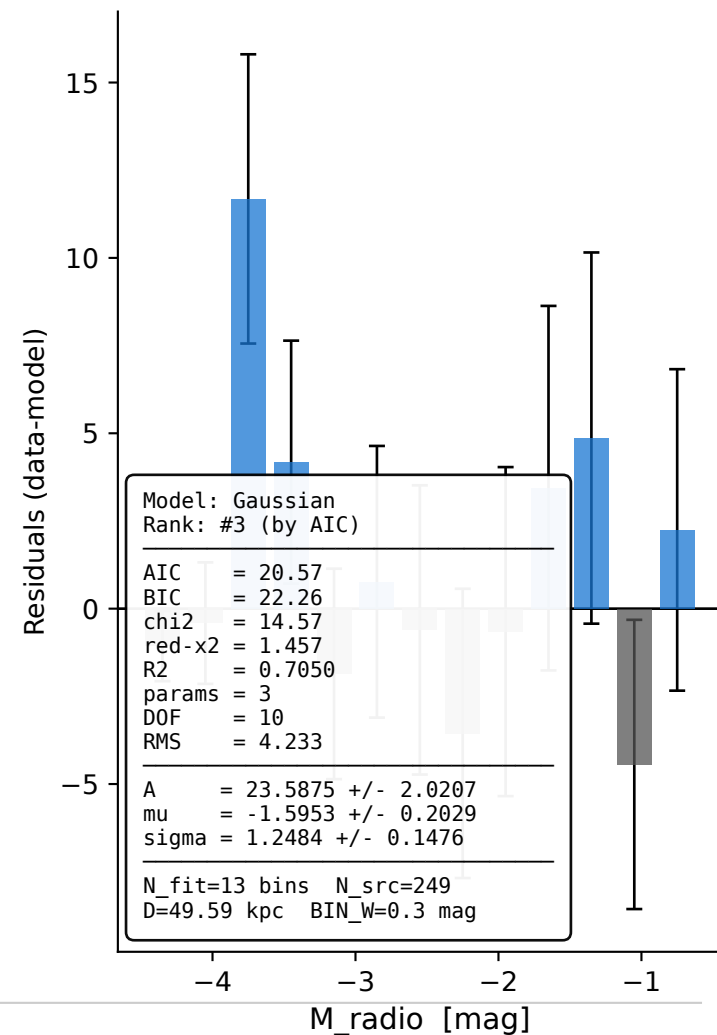
$$N(M) = A * \exp[-0.5 * ((M - \mu) / \sigma)^2]$$

A = amplitude, μ = peak magnitude, σ = width

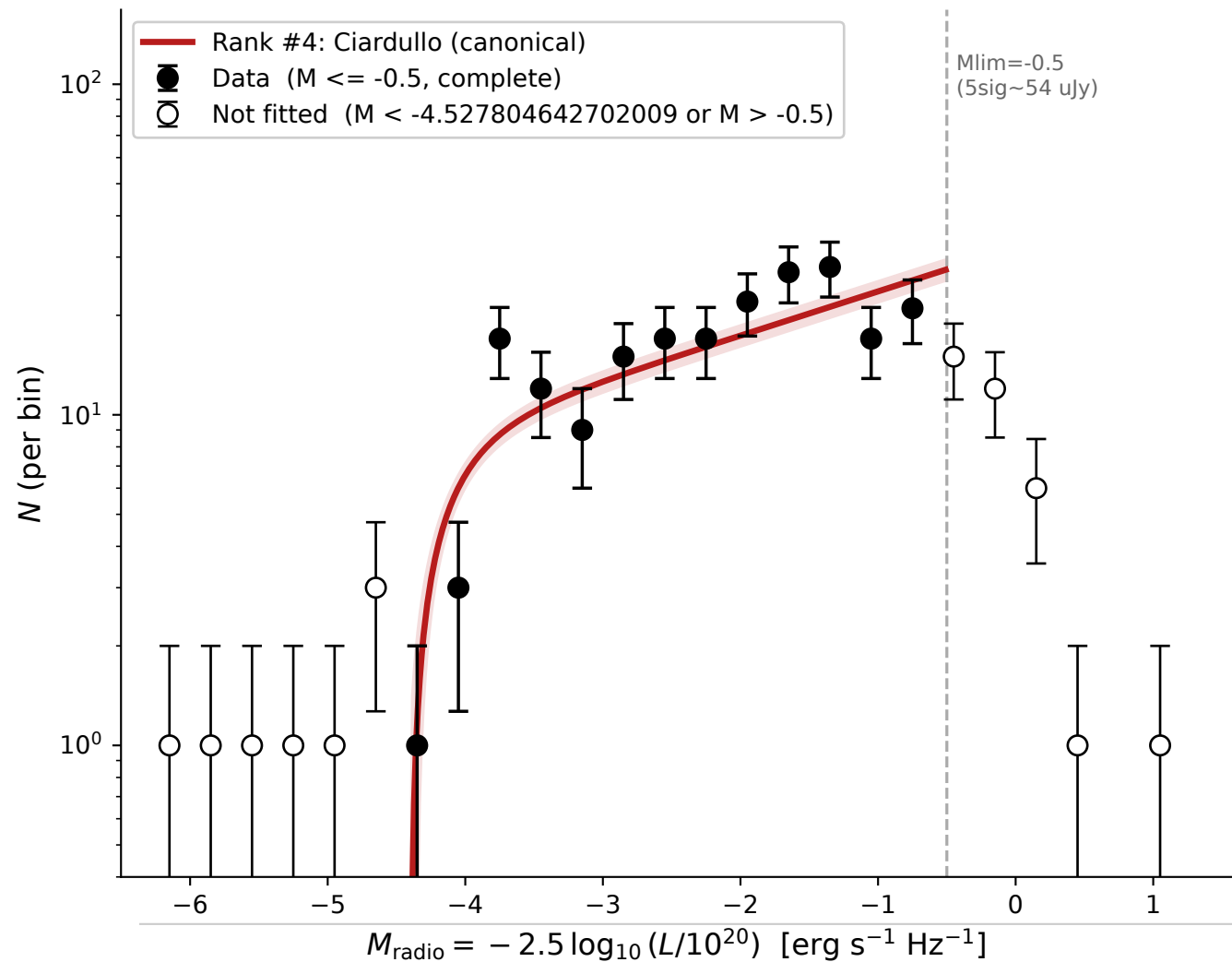
Fitted equation (parameter values substituted):

$$N(M) = 23.59 * \exp[-0.5 * ((M - (-1.595)) / 1.248)^2]$$

Residuals



Rank #4: Ciardullo (canonical)

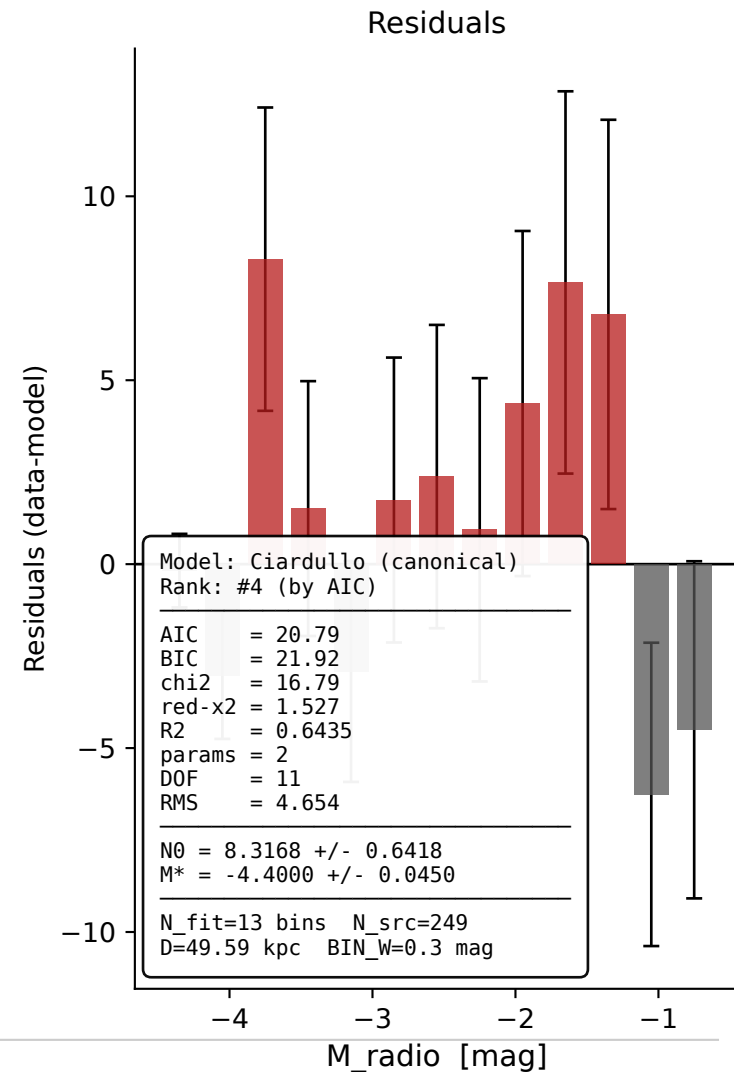


General model equation:

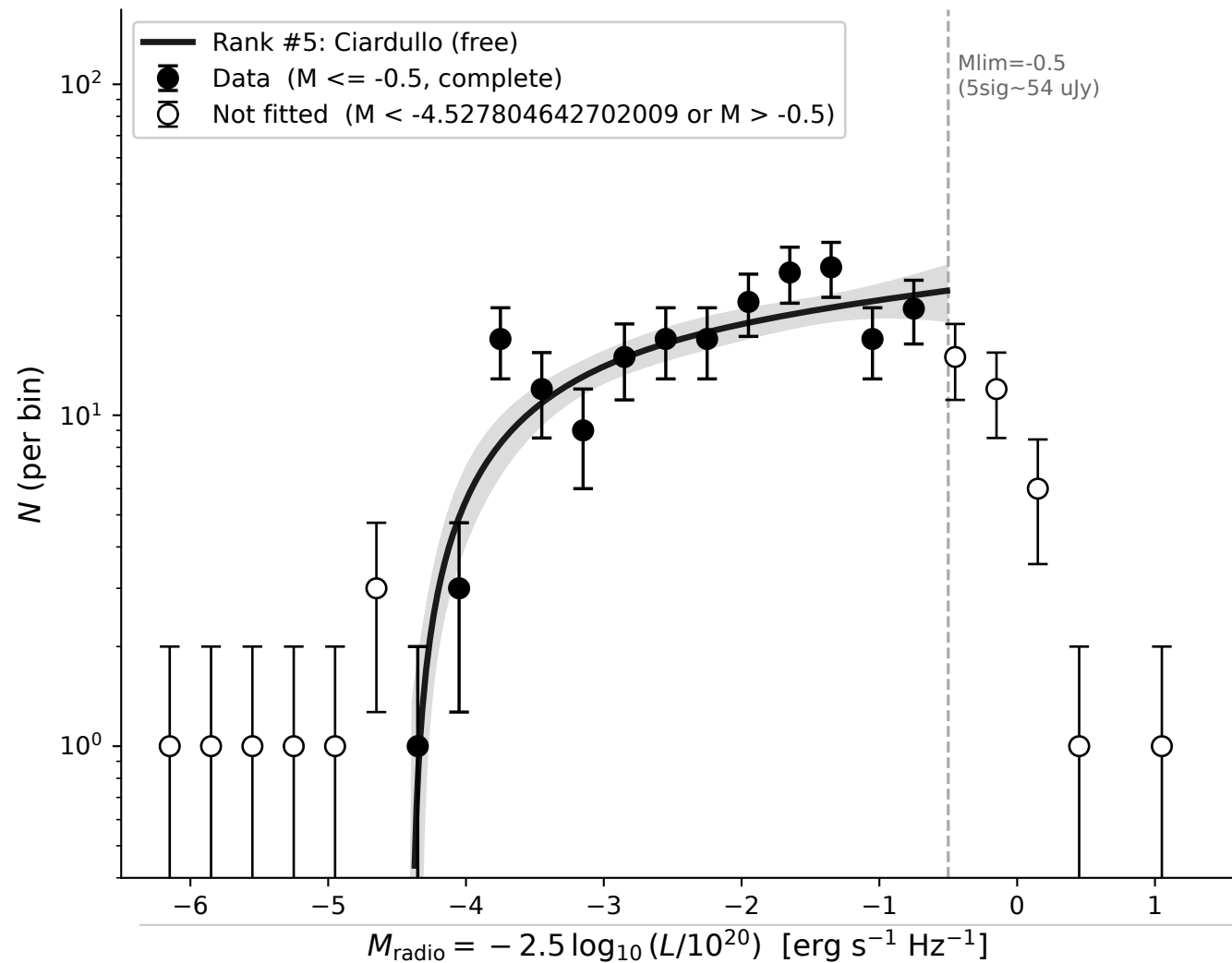
$N(M) = N_0 * \exp(0.307*(M - M^*)) * (1 - \exp(3.0*(M^* - M)))$
 $\alpha = 0.307$, $\beta = 3.0$ (FIXED – Ciardullo 1989 optical values)
 Only N_0 and M^* are free ($k=2$) – most parsimonious Ciardullo form

Fitted equation (parameter values substituted):

$N(M) = 8.32 * \exp(0.307*(M - (-4.400))) * (1 - \exp(3.0*(-4.400 - M)))$
 $N_0=8.32$ $M^*=-4.400$ $\alpha=0.307$ (fixed) $\beta=3.0$ (fixed)
 $L^* = 10^{21.760} \text{ erg/s/Hz}$



Rank #5: Ciardullo (free)



General model equation:

$$N(M) = N_0 * \exp(\alpha * (M - M^*)) * (1 - \exp(\beta * (M^* - M)))$$
 ALL parameters free: N_0 , M^* , α (faint-end slope), β (bright cutoff)
 Ciardullo (1989) form – radio calibration, not fixed to optical values

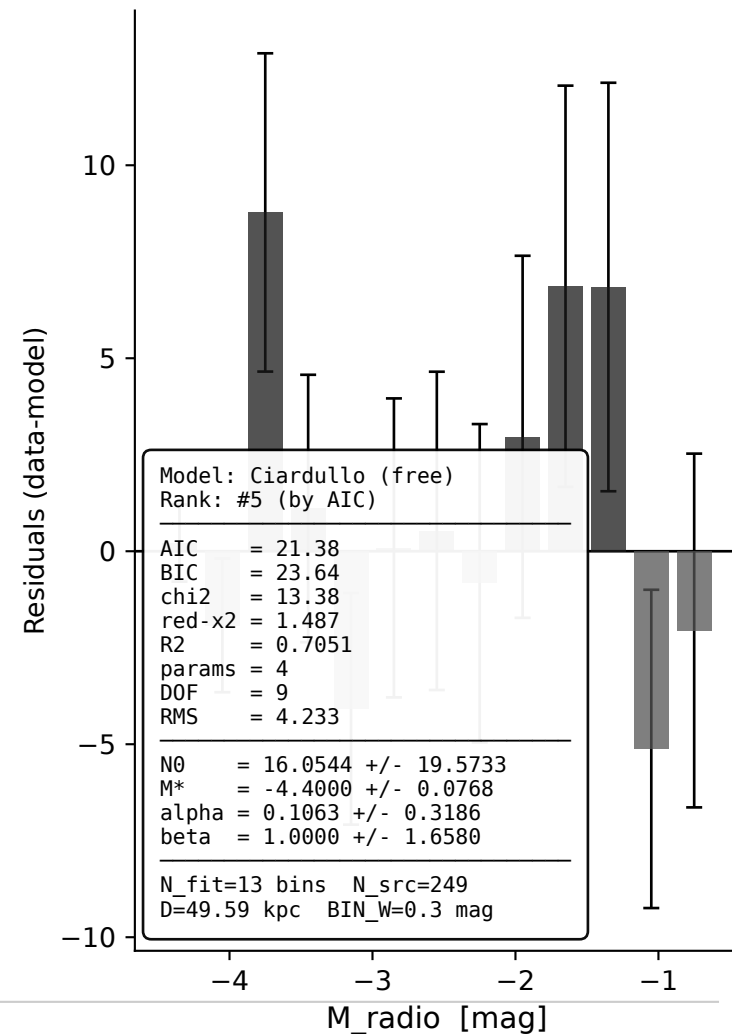
Fitted equation (parameter values substituted):

$$N(M) = 16.05 * \exp(0.106 * (M - (-4.400))) * (1 - \exp(1.000 * (-4.400 - M)))$$

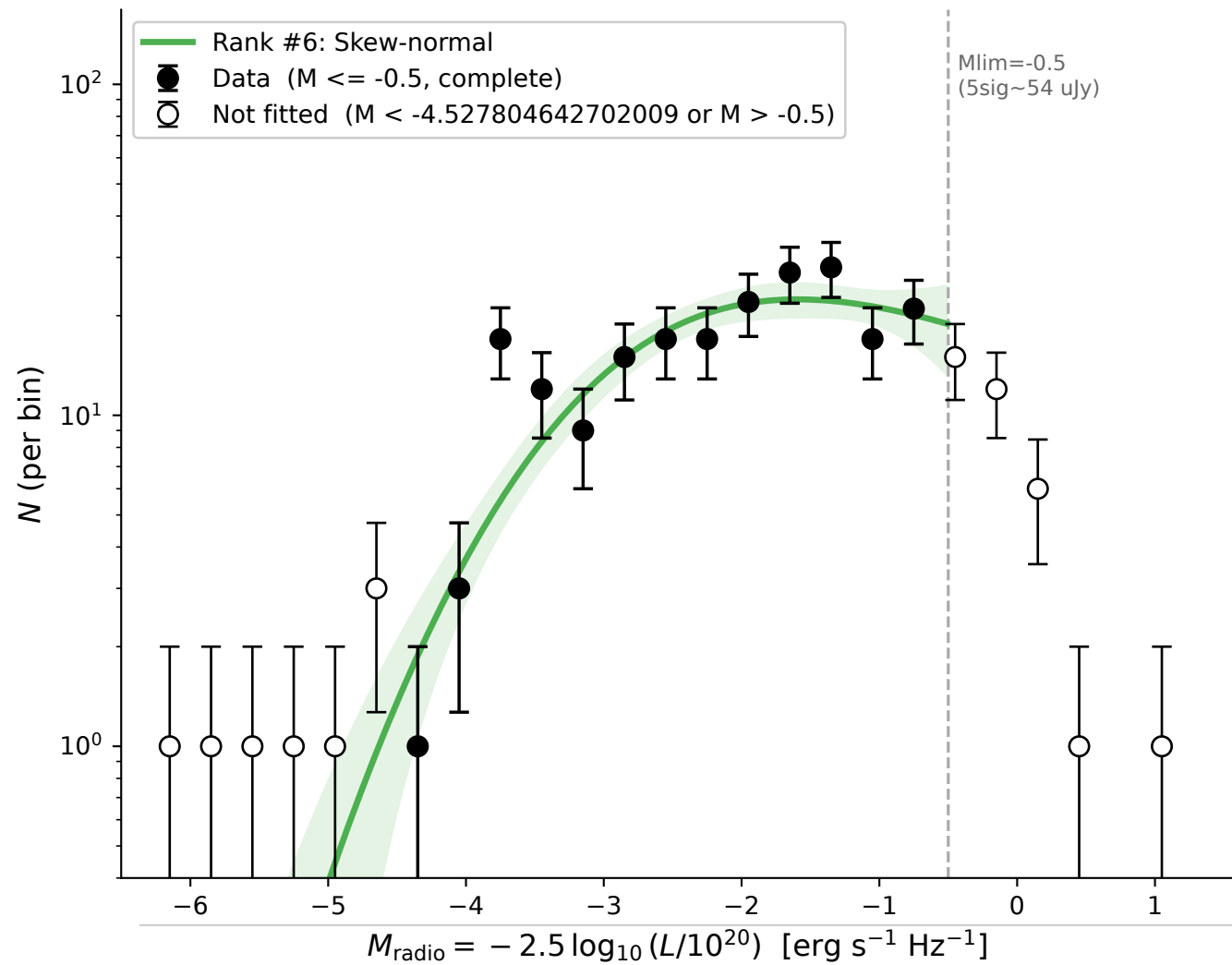
$$N_0=16.05 \quad M^*=-4.400 \quad \alpha=0.106 \quad \beta=1.000$$

$$L^* = 10^{21.760} \text{ erg/s/Hz}$$

Residuals



Rank #6: Skew-normal

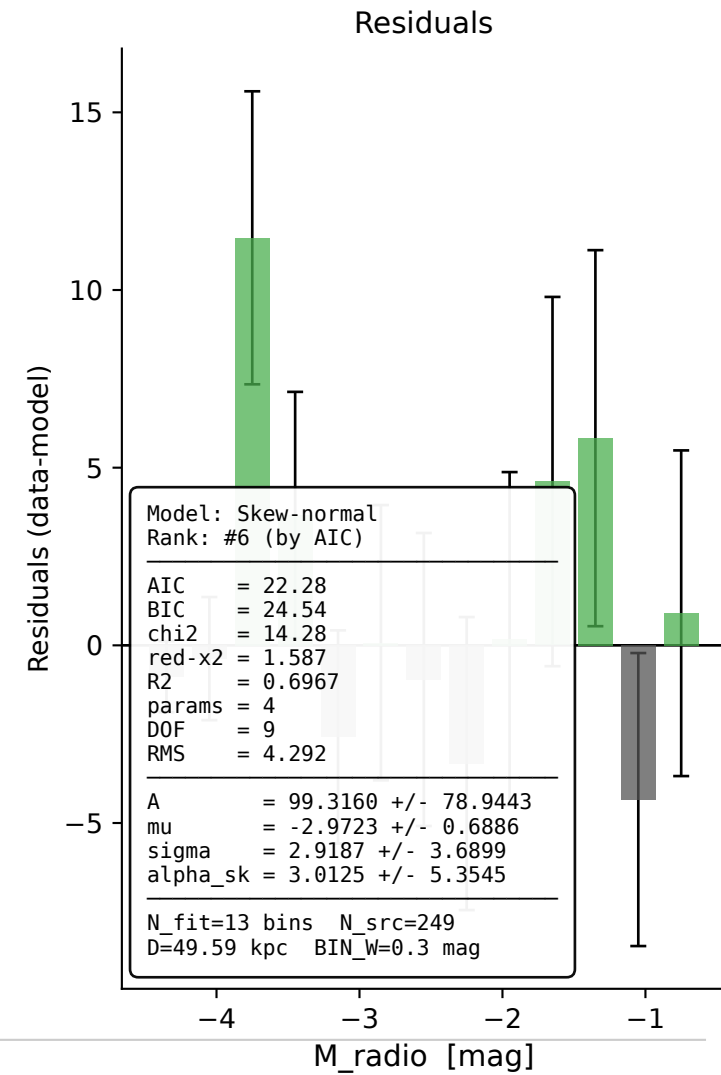


General model equation:

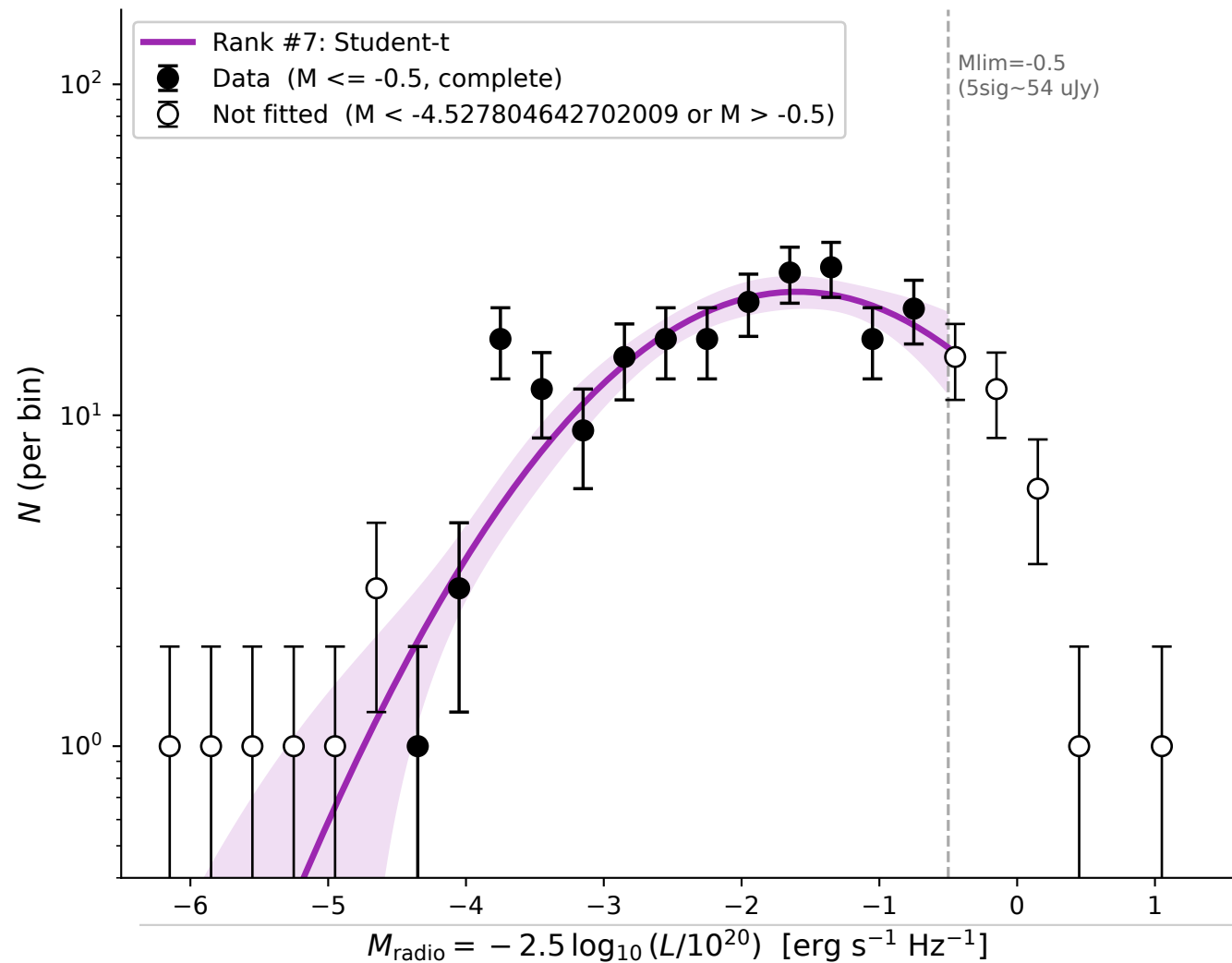
$N(M) = A * f_{\text{SN}}(M; \mu, \sigma, \alpha_{\text{sk}})$
 $f_{\text{SN}}(x) = (2/\sigma) * \phi((x-\mu)/\sigma) * \Phi(\alpha_{\text{sk}} * (x-\mu)/\sigma)$
 ϕ = normal PDF, Φ = normal CDF, $\alpha_{\text{sk}} < 0$ = left-skewed

Fitted equation (parameter values substituted):

$N(M) = 99.32 * f_{\text{SN}}(M; \mu=-2.972, \sigma=2.919, \alpha_{\text{sk}}=3.013)$



Rank #7: Student-t



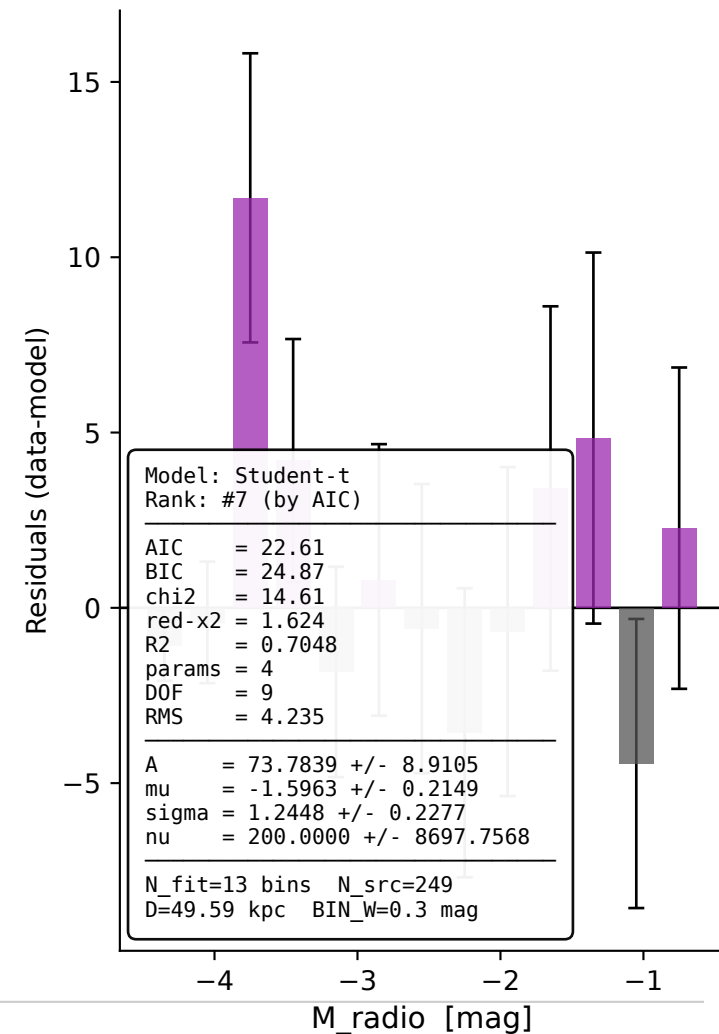
General model equation:

$N(M) = A * t_{\text{nu}}((M - \mu) / \sigma)$
 $t_{\text{nu}}(x)$ proportional to $(1 + x^2/\nu)^{-(\nu+1)/2}$
 A = amplitude, μ = centre, σ = scale, ν = degrees of freedom

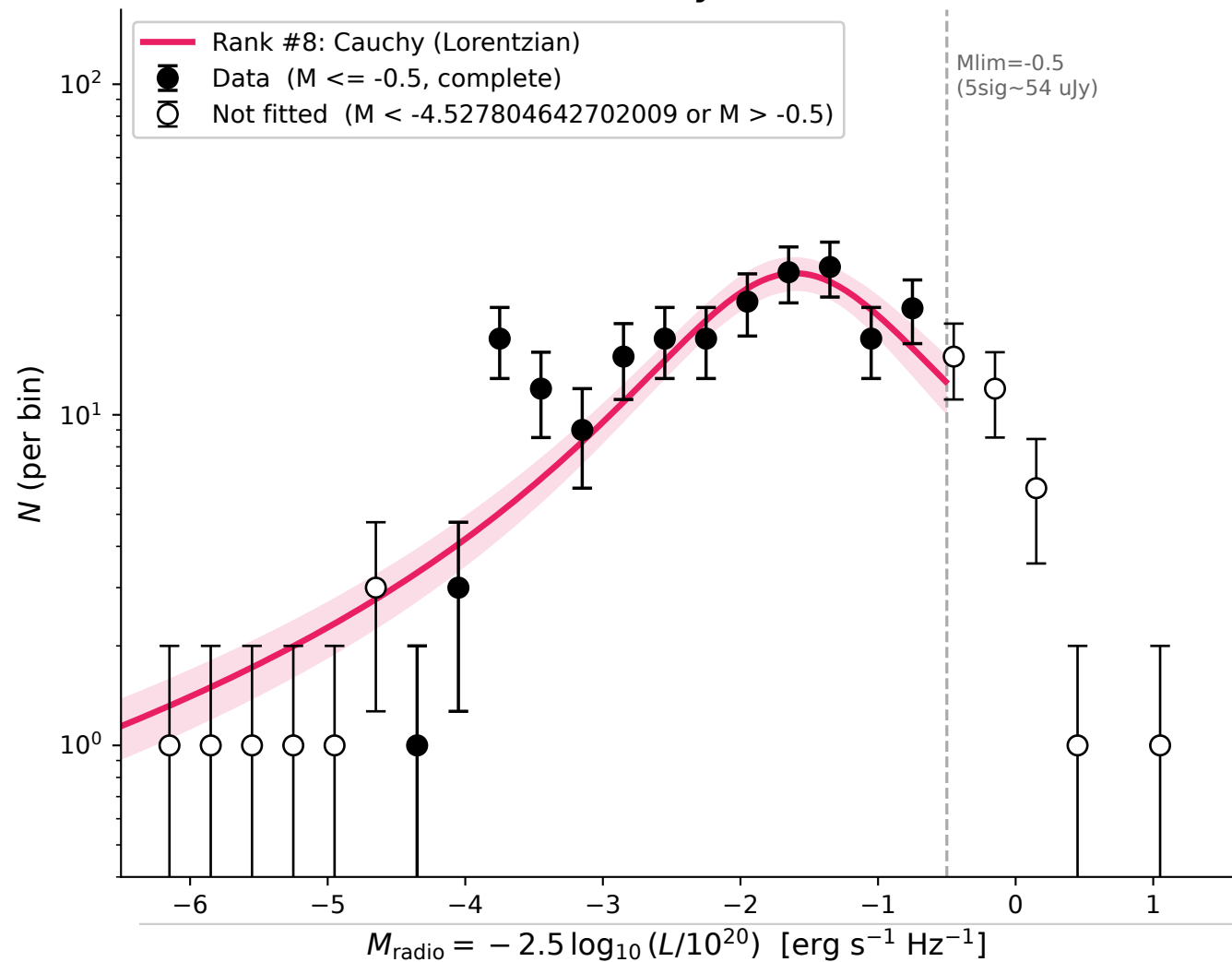
Fitted equation (parameter values substituted):

$N(M) = 73.78 * t_{200.0}((M - (-1.596)) / 1.245)$

Residuals



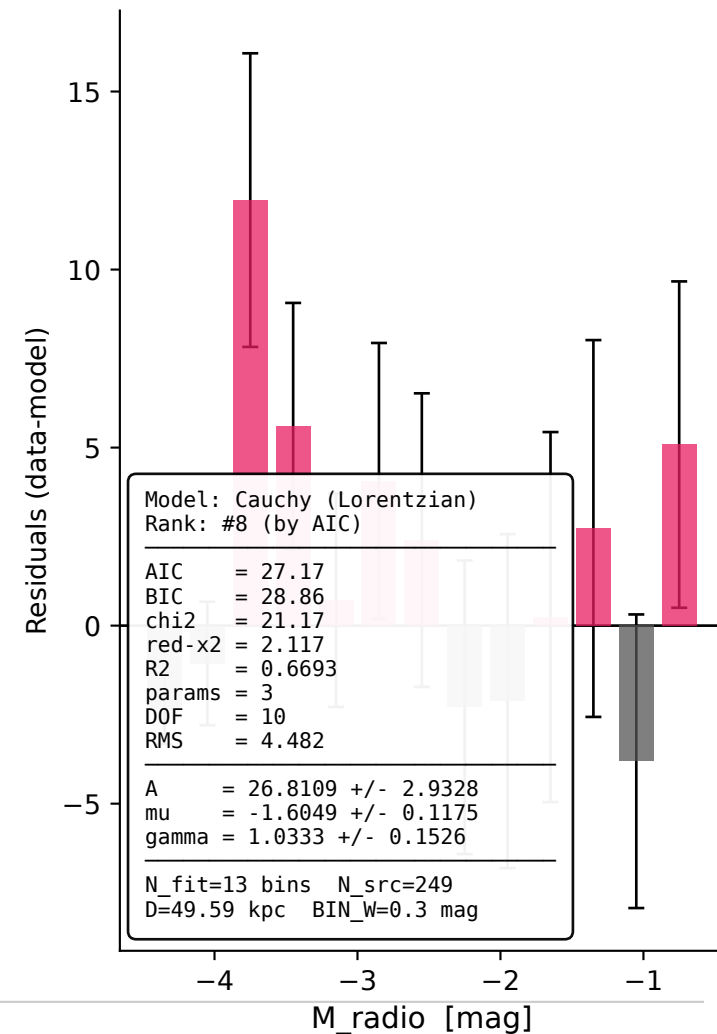
Rank #8: Cauchy (Lorentzian)



General model equation:

$$N(M) = A / [1 + ((M - \mu) / \gamma)^2]$$
 Heavy-tailed symmetric distribution (heavier tails than Gaussian)
 A = peak height, μ = centre, γ = half-width at half-maximum

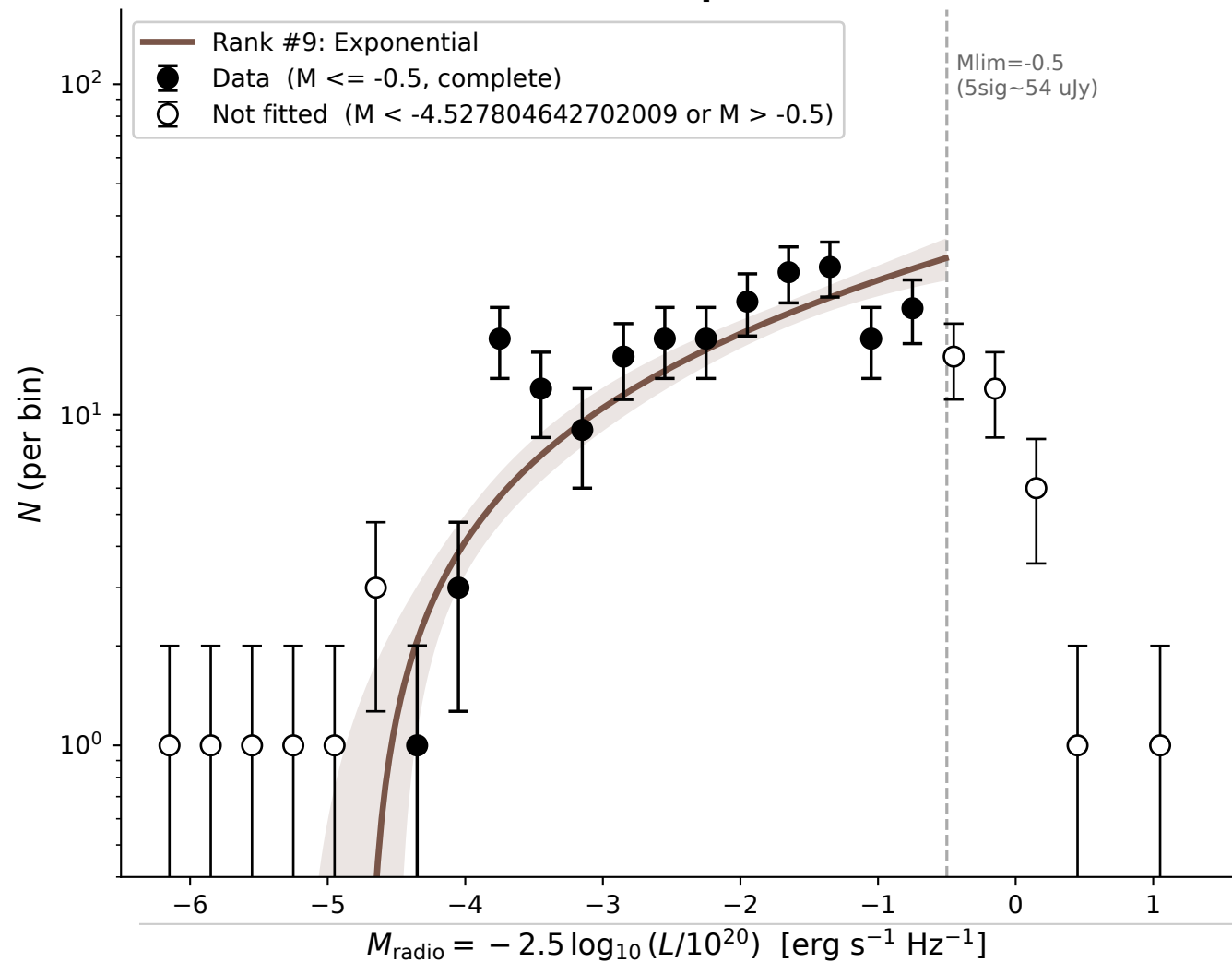
Residuals



Fitted equation (parameter values substituted):

$$N(M) = 26.81 / [1 + ((M - (-1.605)) / 1.033)^2]$$

Rank #9: Exponential

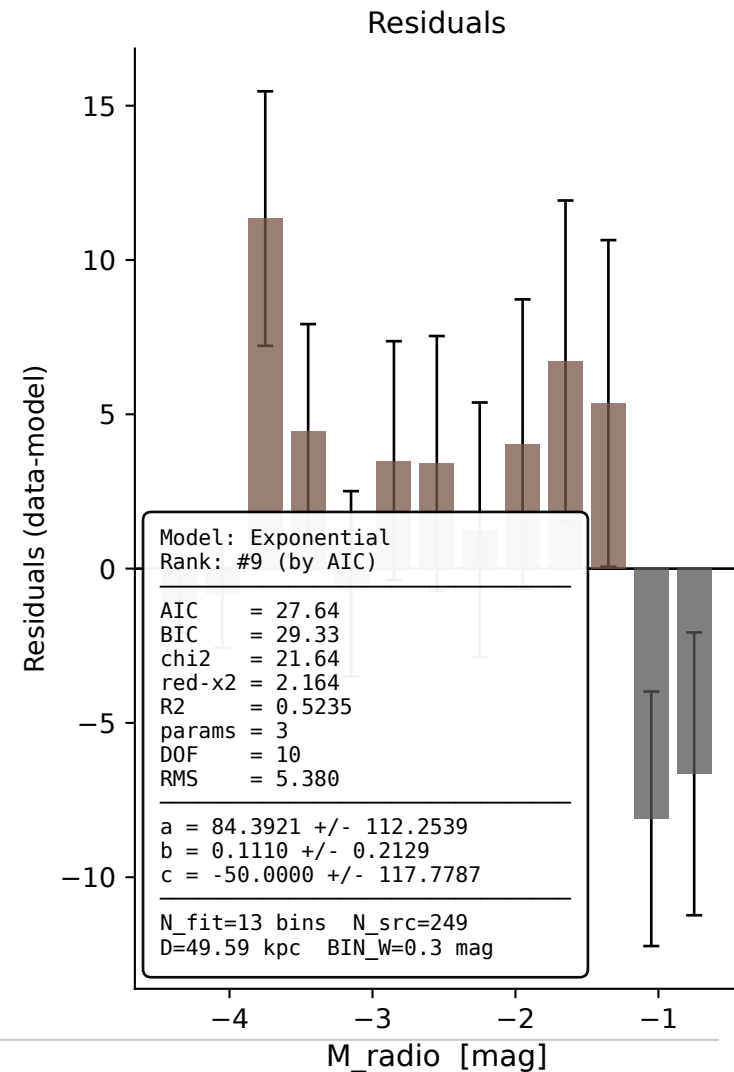


General model equation:

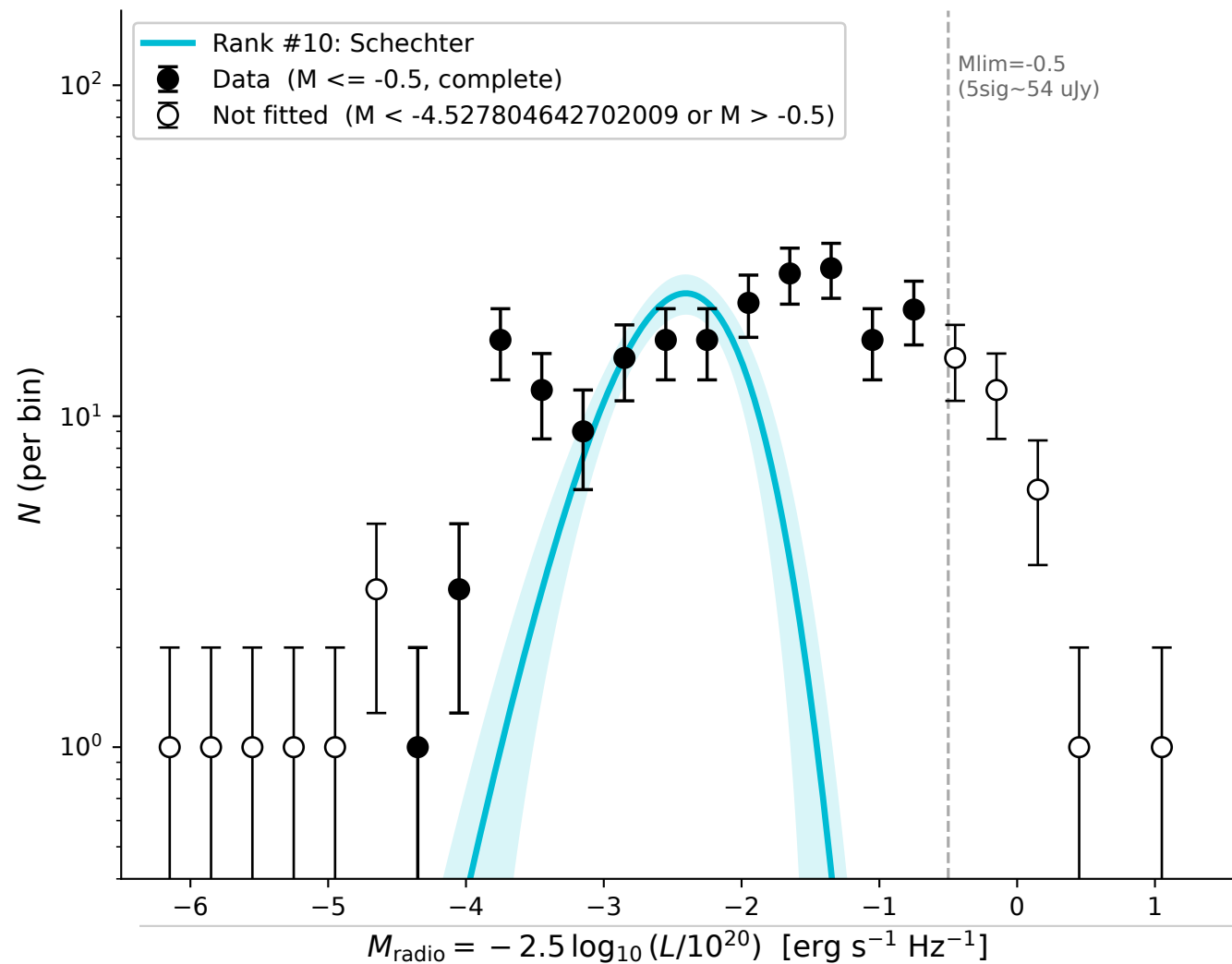
$N(M) = a * \exp(b * M) + c$
 Simple exponential rise toward faint magnitudes
 a = amplitude, b = rate (>0 = rising toward faint), c = offset

Fitted equation (parameter values substituted):

$N(M) = 84.3921 * \exp(0.1110 * M) - 50.0000$



Rank #10: Schechter



General model equation:

$$N(M) = \phi * 10^{(0.4 * (\alpha + 1) * (M - M^*))} * \exp[-10^{(0.4 * (M - M^*))}]$$
 ϕ = normalisation, M^* = bright cutoff, α = faint-end slope
 Originally for galaxy luminosity functions (Schechter 1976)

Fitted equation (parameter values substituted):

$$N(M) = 0.203 * 10^{(0.4 * (5.000 + 1) * (M - (-4.350)))} * \exp[-10^{(0.4 * (M - (-4.350)))}]$$

Residuals

